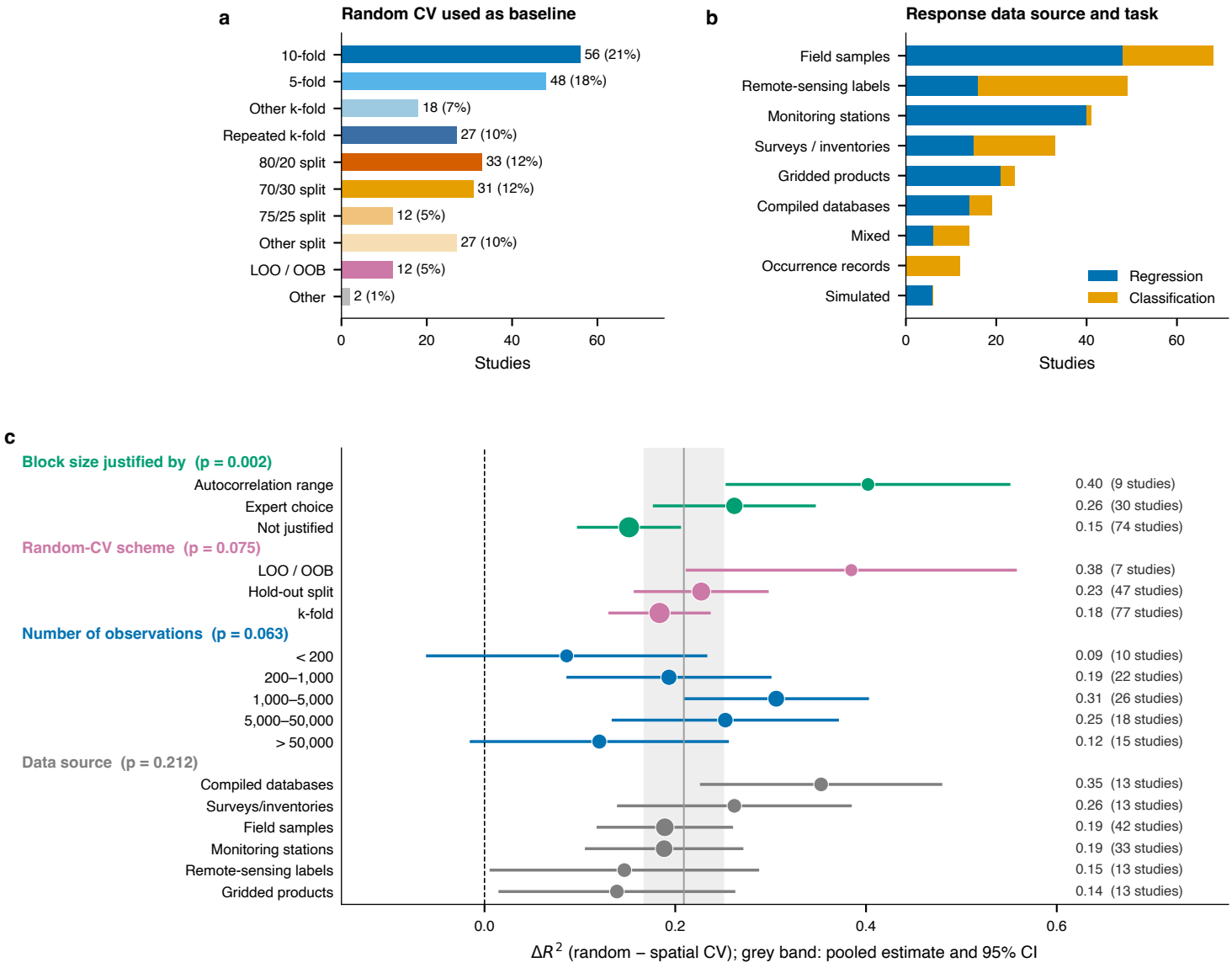
a
Random CV used as baseline
10-fold
5-fold
Other k-fold
Repeated k-fold
80/20 split
70/30 split
75/25 split
Other split
LOO / OOB
Other
56 (21%)
48 (18%)
18 (7%)
27 (10%)
33 (12%)
31 (12%)
12 (5%)
27 (10%)
12 (5%)
2 (1%)
0
20
40
60
Studies
b
Response data source and task
Field samples
Remote-sensing labels
Monitoring stations
Surveys / inventories
Gridded products
Compiled databases
Mixed
Occurrence records
Simulated
Regression
Classification
0
20
40
60
Studies
c
Block size justified by (p = 0.002)
Autocorrelation range
Expert choice
Not justified
Random-CV scheme (p = 0.075)
LOO / OOB
Hold-out split
k-fold
Number of observations (p = 0.063)
< 200
200–1,000
1,000–5,000
5,000–50,000
> 50,000
Data source (p = 0.212)
Compiled databases
Surveys/inventories
Field samples
Monitoring stations
Remote-sensing labels
Gridded products
0.40 (9 studies)
0.26 (30 studies)
0.15 (74 studies)
0.38 (7 studies)
0.23 (47 studies)
0.18 (77 studies)
0.09 (10 studies)
0.19 (22 studies)
0.31 (26 studies)
0.25 (18 studies)
0.12 (15 studies)
0.35 (13 studies)
0.26 (13 studies)
0.19 (42 studies)
0.19 (33 studies)
0.15 (13 studies)
0.14 (13 studies)
0.0
0.2
0.4
0.6
ΔR² (random – spatial CV); grey band: pooled estimate and 95% CI